A preliminary study on indicator framework for enterprise, based on COBIT 5 processes and SOA approach

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Abstract—**Organizations become dependent on Information Technology (IT) to fulfill their corporate aims, meet their business needs and deliver value to customers. For effective and efficient utilization of IT, we intend to contribute to the alignment between IT and organizational strategies. The challenge is to make the enterprise and its Information System (IS) as responsive as possible within the regard to changes in enterprise while taking into account the enterprise operational performance. When a change happens or a new opportunity comes, organizations do not know which assets are linked to which business processes and which services, especially IT services, to bring up first and which can wait until later. Our principal aim is to propose an indicator framework to track and control business-IT alignment. However, this paper is a preliminary study on this framework. We are inspired by COBIT 5 processes as indicators and believe that the Service-Oriented Architecture (SOA) applied to these processes interlinks and can interact well the different processes. The validity and applicability of our theoretical study will be evaluated for future work.**

Service-oriented architecture (SOA); key performance indicator (KPI); indicator; COBIT processes; alignment

I. Introduction

Since these recent decades, information and communication technologies have become essentials to human resources development in sociocultural and economic terms, particularly to the survival of the business owned many sectors. The use of IT evolved considerably, providing assistance to the enterprise in its operational and strategic management. Organizations become dependent on IT to fulfill their corporate aims, meet their business needs and deliver value to customers [1]. However, there are a lot of challenging issues that they face in its utilization. Among these is the fact that traditional IS organization strategies are designed to withstand in relatively stable environments where changes in the enterprise's context are limited [2]. The problem is to make the enterprise and its IS as responsive as possible with the regard to changes in enterprise while taking into account the enterprise operational performance. When a change happens or a new opportunity comes, organizations do not know which assets are linked to which business processes and which services, especially IT services, to bring up first and which can wait until later. To address this challenge in order to have effective and efficient utilization of IT in enterprise, we propose a preliminary study on indicator framework that will allow us to measure the control capabilities and track alignment between IT and business strategies.

The rest of this paper is organized as follows. In section II, we present an overview of indicator. We are inspired by COBIT 5 processes for our indicators, thus, section III makes accent on COBIT 5 processes and areas of activity. In section IV, we explore how SOA approach can be applied to these processes and so that the different processes are interlinked and can interact well in order to adapt to structural changes. Section V discusses and relates our work to other works that have been examined during our research. We conclude our paper and lists the future research directions in section VI.

II. Indicator: an overview

A. *Definitions and general concepts*

The term indicator, from Latin indicator, qualifies that is the culprit [3]. An indicator is a series of observations about a specific variable, or set of variables, believed to represent the behavior of some specific occurrence [4]. Indicators are typically relied upon to monitor and analyze changes in important variables, and they most often take form of a quantitative index. On the one hand, they are developed to analyze and to forecast the turning points of business cycles. On the other hand, they are developed on the basis of regression equations to predict future values of time series [5]. An indicator can be classified according to its destination: monitoring, analysis or forecasting [3].

Indicators are tools that measure, simplify and communicate important issues and trends, and are used throughout society in a multitude of ways. They can be used to translate and communicate complex information into easily understandable units [6]. Three basic functions that they provide are [7]: (1) control – they enable managers and workers to evaluate and control the performance of the resources which they are responsible; (2) communication – they communicate performance not only to internal workers and managers for a purposes of control, but also to external stakeholders for other purposes; and (3) improvement – they identify gaps (between performance and expectation) that ideally point the way for intervention and improvement.

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B. *Terminology*

Bellow we give meanings of some terms can be found in the literature on indicator context at enterprise level. The performance in each indicator is measured, it is multiplied by its importance, usually expressed as weights, and the results are summed over all indicators to yield an aggregated measure for comparison [8]. An index number can be formally defined as a number expressing the value of some entity, say price or gross national product, at a given period of time in absolute number form but related to a base period which is arbitrary set equal to 100 [4].

Performance indicator (PI) or key performance indicators (KPIs) provide the most important performance information that enables organizations or their stakeholders to understand whether the organization is on track or not [9]. They are commonly used by an organization to measure, quantify and evaluate its success or the success of a particular activity in which it is engaged.

In Table 1, Samsonowa [10] defines measures, metric, PI and KPI as the follow:

TABLE I. DEFINITIONS: MEASURE, METRIC, PI, KPI

Term	Definition
Measure	A quantifying value
Metric	A metric puts a measure into a certain context. The context is given by an item or an object or a set of items or objects. It defines a unit of measure and a reference unit
PI	A performance indicator is an auxiliary metric that partially reflects the performance of an organizational unit
KPI	KPIs are a set of performance indicators that are selected upfront and agreed on by management to be the most representative and/or critical performance indicators. A KPI is an element of this set

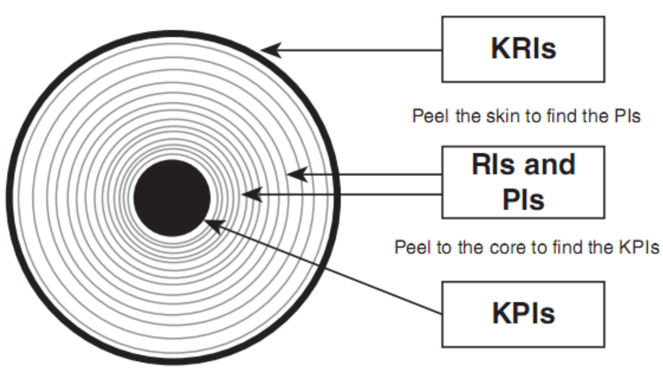

Figure 1. Four type of performance measure [11].

Parementer [11] states four types of performance measures: (1) key result indicators (KRIs) tell you how you have done in a perspective or critical success factor; (2) result indicators (RIs) tell you what you have done; (3) performance indicators (PIs) tell you what to do, and (4) KPIs tell you what to do to increase performance dramatically. The author uses an onion analogy to describe the relationship between these four measures, as shown in Fig. 1. The outside skin describes the overall condition of the onion, which is a KRI. However, as we peel the layers off the onion, we find more information. The layers represent the various performance and result indicators, and the core represents the KPI. In addition, KPIs serve to reduce the complex nature of organizational performance to a small, manageable number of key indicators that provide evidence that can in turn assist decision making and ultimately improve performance [12].

C. *Indicators construction and framework*

An indicator is reliable depends primarily on two factors: the validity of the methodology and the quality of the data [4]. Generally, each indicator refers to a specific target, that is to say a sort of reference point used as a basis of comparison [7].

Franceschini, Galetto and Maisano [7] assume that indicators selection should be performed considering: quality policy, quality targets, the area of interest, performance factors, process targets. They describe some basic requirements to define indicators, which are: representativeness, simple and easy to interpret, capable to indicate time-trends, sensitive to changes within or outside the organization, easy data collecting and processing, easy and quick to update.

A framework organizes how combinations of indicators are selected and used [13]. Indicators are usually bundled together and linked to other information and management tools to serve an overall assessment purpose. A framework generally provides an outer boundary as well as an internal structure for deriving and using information. There are two ways to understand or visualize a "frame": as a "picture frame" that focuses attention on what is happening inside the frame rather than on the walls outside; or as a "building frame," like a set of steel bars, locking various elements of a house into their proper position. Each of these notions suggests a different perspective on the importance of frames, namely the focus (external frame) and organization (internal frame) of information, respectively. Frameworks usually do both to some degree, i.e., provide essential information and a context for planning and decision-making. We believe that framework is a useful tool for strategic alignment as well as the use of indicators.

III. COBIT 5: A BRIEF DESCRIPTION

A. *Reasons for choosing COBIT 5*

COBIT (Control Objectives for Information and Related Technology) is developed by Information Systems Audit and Control Association (ISACA). Five major versions are released and COBIT 5 is the last one [14, 15]. COBIT 5 was created to directly address business needs, and it cross-references to all relevant, internationally recognized standards and frameworks. In driving effective governance of enterprise IT through the use of COBIT 5, enterprises are likely to realize the following benefits [16]: (1) better value creation through effective and innovative use of enterprise IT; (2) increased business-user satisfaction with IT engagement and services; (3) increase compliance with

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relevant laws, regulations, and policies; (4) improved relationship between business needs and IT objectives; and (5) increased financial return from governance of enterprise IT by obtaining the greatest value from investment in technology.

COBIT offers effective practices throughout a framework and lays down activities in an organized and flexible structure [17]. These practices assist in optimizing IT-enabled investments, guarantee delivery of service and offer protection against who's accountable for the wrongs. COBIT 5 is generic and useful for enterprises of all sizes [18].

B. COBIT 5 process reference model

COBIT 5 process reference model defines and describes in detail a number of governance and management processes. It represents all of the process normally found in an enterprise relating to IT activities, providing a common reference model understandable to operational IT and business managers [19].

Enterprises implement governance and management processes such that the key areas are covered, as shown in Fig. 2. The difference between types of processes lies within the objectives of the processes. Governance processes deal with the stakeholder governance objectives and include practices and activities aimed and evaluating strategic options, providing direction to IT and monitoring the outcome. Management processes cover the responsibility areas of plan, build, run and monitor (PBRM) enterprise IT, and they provide end-to-end coverage of IT [19]. 37 IT processes spread over governance and management domains [20], which are divided into domain process [18]: Evaluate, Direct and Monitor (EDM); Align, plan and organize (APO); Build, acquire and implement (BAI); Deliver, service and support (DSS); and Monitor, Evaluate and Assesses (MEA).

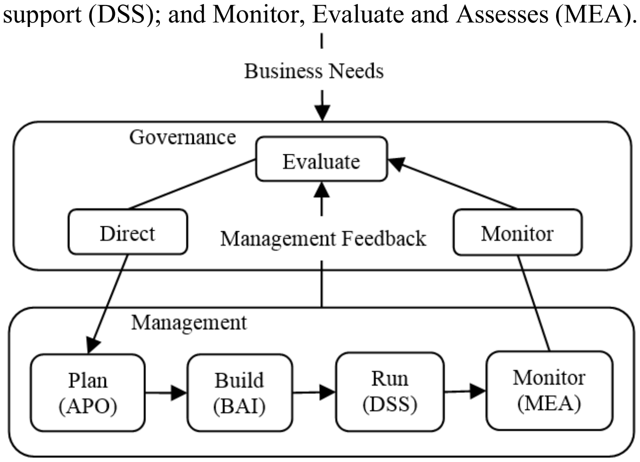

Figure 2. COBIT 5 governance and management key areas [19].

C. Process assessment

COBIT 5 product set includes a process capability model, based on the internationally recognized ISO/IEC 15504 Software Engineering – Process Assessment standard. This model will achieve the same overall objectives of process assessment and process improvement support, it will provide a means to measure the performance of any of the governance (EDM-based) processes or management (PBRM-based) processes, and will allow areas for improvement to be identified [19]. The COBIT 5 process capability approach can be summarized as shown in Fig. 3.

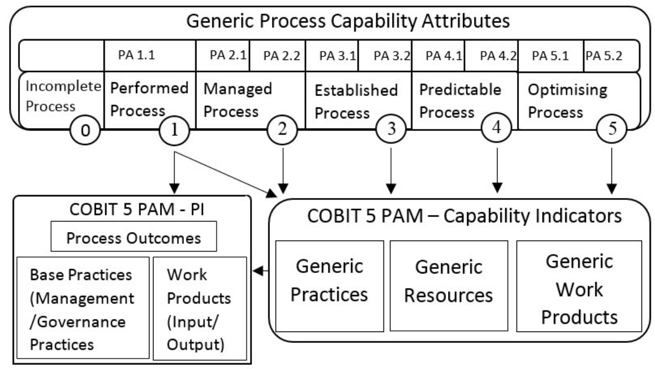

Figure 3. Summary of the COBIT 5 process capability model [19].

There are six levels of capability that a process can achieve, including an "incomplete process" designation if the practices in it do not achieve the intended purpose of the process: 0 – Incomplete process; 1 – Performed process (performance attribute (PA) 1.1 process performance); 2 – Managed process (PA 2.1 performance management and PA 2.2 work product management); 3 – Established process (PA 3.1 process definition and PA 3.2 process deployment); 4 – Predictable process (PA 4.1 process management and PA 4.2 process control); and 5 – Optimizing process (PA 5.1 process innovation and PA 5.2 process optimization). Each capability level can be achieved only when the level below has been fully achieved.

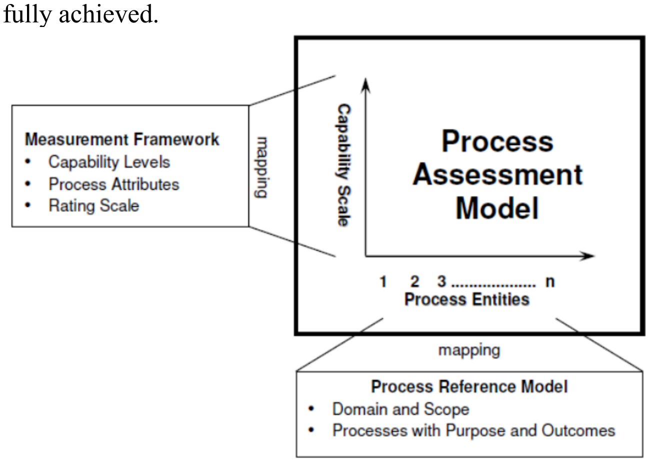

Figure 4. Components of ISO/IEC 15504 Process Assessment [21].

As Fig. 4 presents, the Process Assessment Model (PAM) defines a two-dimensional model of process capability [21]. In one dimension, the process dimension, the processes are defined and classified into process categories. In the other dimension, the capability dimension, a set of process attributes grouped into capability levels is defined. The model expands upon the Process Reference Model by adding the definition and use of assessment indicators.

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Assessment indicators comprise indicators of process performance and process capability and are defined to support an assessor's judgment of the performance and capability of an implemented process. The process attributes provide the measurable characteristics of process capability. They are evaluated on a four point ordinal scale of achievement, as defined in ISO/IEC 15504-2 [21]: N – Not achieved (0 – 15% achievement); P – Partially achieved (15 – 50% achievement); L – Largely achieved (50 – 85% achievement); and F – Fully achieved (85 – 100% achievement). They provide insight into the specific aspects of process capability required to support process improvement and capability determination.

IV. SOA APPROACH

A. *Definitions and reasons for choosing SOA*

There is no single, universally definition of a SOA. A definition aggregated from different authors and organizations is [22]: "*SOA is an application framework that takes heterogeneous, distributed, complex and inflexible business applications and breaks them down into individual business functions, called services. A SOA is able to build, deploy, discover, integrate and reuse these services independent of applications and the computing platforms on which they run, making business processes simpler, more consistent and flexible.*"

Literature review further confirmed that SOA is a service-based architecture which defines services and the methods of communication between them [23], an architectural blueprint, a way of developing applications, and set of best practices [24] designed around loosely coupled software components called services, which can be orchestrated to improve business agility [25].

SOA establishes an architectural model that aims to enhance the efficiency, agility, and productivity of an enterprise by positioning services as the primary means through which solution logic is represented in support of the realization of strategic goals associated with service-oriented computing [26]. It is a widely accepted approach for wrapping and integrating legacy systems in terms of supporting software reusability and scalability as well as business flexibility and cost-reduction for IT development/ maintenance [22].

SOA also promises the ability to improve the alignment of business and IT through loose coupling and design around services [25]. It can better align IT initiatives with business requirements [27]. It will benefit organizations in different ways, depending on their respective goals and the manner in which SOA and its supporting cast of products and technologies is applied [28]. SOA is expected to increase empowerment of both business and IT communities.

B. *SOA concept*

SOA is based on several principles, such as loose-coupling, reuse, explicit boundaries, shared contracts and schemas, and meta-model [29]. These principles essentially involve orienting applications in the enterprise towards providing services, and consuming these services in simple flowchart like business processes that are modeled very closely on the enterprise business processing [30].

A service is an act or performance offered by one party to another [31]. It is a connection to an application, providing access to certain of its functionalities [32]. The functions provided by a service can be treatments, information researches. A way to describe a service is a unit of work done by a service provider to achieve desired results for a service consumer [33].

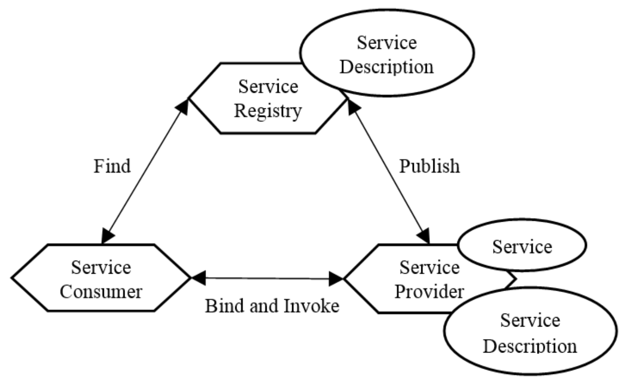

Figure 5. Collaborations of services in an SOA [33].

The collaborations of services in an SOA, as shown in Fig. 5, follow the "find, bind and invoke" paradigm where a service consumer performs dynamic service location by querying the service registry for a service that matches its criteria. If the service exists, the registry provides the consumer with the interface contract and the endpoint address for the service [33]. A service description in its most basic format establishes the name of the service and the data expected and returned by the service. The manner in which services use service descriptions results in a relationship classified as loosely coupled [28].

Services are consisted of functionality, reachability contract and policy, action model and process model semantics and structure [33]. They also are considered as means of urbanized IS integration since they enable urbanized applications to communicate and work together effectively [31]. In service-oriented solutions, each task can involve any number of services. The interaction of a group of services working together to complete a task can be referred to as a service activity. In SOA, activities represent any service interaction required to complete business tasks. The scope of a service activity is primarily concerned with the processing and communication between services only. An activity is a generic concept used to represent a task or a unit of work performed by a set of services [28].

Services are identified in two possible ways [29], depending on the development approach used for services: top-down approach – services are identified through the business process modelling, bottom-up approach – services are identified based on the functional specification of the service. Middle-out approach has both push up into the enterprise scope and, at the same time, push down into immediate deliverables. Business Process Management

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(BPM) can drive the discovery and design of required services [34].

C. SOA model

Rosen, Lublinsky, Smith and Balcer [34] affirm that the SOA architectural approach needs: (1) a business architecture that lays out a roadmap for the processes and services of the enterprise now and overtime, and identifies the functional and application capabilities to support those services; (2) an information architecture that lays out a roadmap for the shared enterprise semantics and data model; (3) an application architecture that defines a hierarchy of service types; and (4) a technology architecture that defines what the technologies are and how they are used to support processes, services, integration, data access and transformations, and so on. According to authors, SOA reference architecture should: (1) support enterprise concepts, particularly the subarchitectures of business, information, application, and technology; (2) specify a hierarchy and taxonomy of services and service types; (3) define how services fit into an overall enterprise application; (4) provide a separation between business, application, and technology concepts; and (5) be integrated into the development process. Fig. 6 shows the major components of an SOA reference architecture.

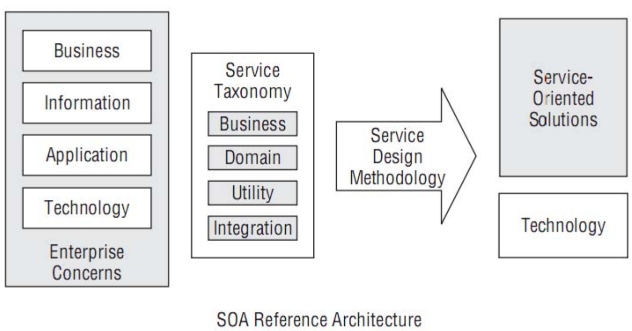

Figure 6. Aspect of an enterprise SOA reference architecture [34].

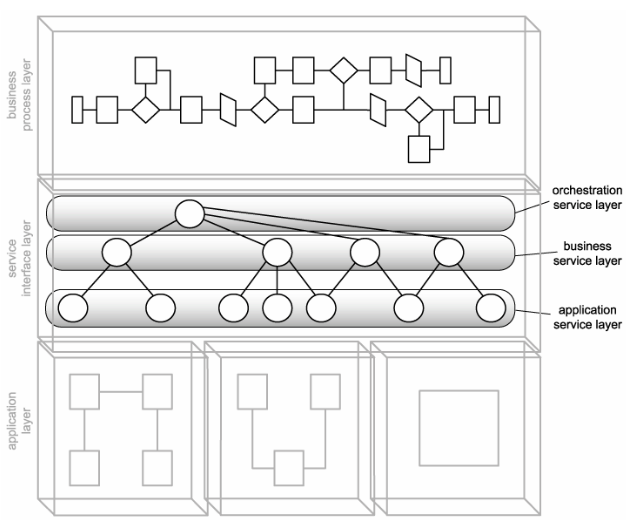

Figure 7. Common service layers [28].

A service layer is an abstraction level which is introduced in order to structure services [22]. Erl [28] locates the service interface layer between the business process and application layers. Author identify three layers of abstraction for SOA (see Fig. 7): the application service layer, the business service layer, and the orchestration service layer

V. Related work

Several works that relate to KPI framework, COBIT and SOA have been examined during our research. Several relevant aspects are incorporated to our research.

Ganesan and Paturi [35] state business strategies and the underlying business objectives have to be tracked and monitored for performance so that can help to make informed decisions. They detail importance of KPIs linking business strategy to business processes, business role players and business products and services by defining a fitment between Composite Enterprise Business Architecture Framework with Business Motivation Model of Business Rules Group and Balanced Scorecard approach. They attempt to define a framework for KPI classification and relate it to business process hierarchy. They prescribe KPI cycle approach for effectively utilizing the Composite KPI framework and classification schema.

Hojaji and Shirazi [36-38] propose a SOA governance framework based on COBIT 4.1. Their framework is obtained by enforcing governance structures of COBIT and applying service management activities into a lifecycle approach. It is also obtained by extending characteristics of COBIT and applying ITIL service lifecycle activities to support the SOA governance principles and requirements. They considered COBIT as a comprehensive IT governance framework in their comparison to see whether it supports the SOA governance elements or not. However, SOA governance is essentially different from IT governance, COBIT is a well-accepted framework that has been implemented by many companies, and is based on IT service management approach.

Gu and Zhang [27] introduce a SOA based Enterprise Application Integration approach. They model the business process of the enterprise and identify process blocks that can be group as services. They highlight three different service categories to encapsulate the enterprise business process elements: adapter services, composite services, process services. They aim to provide a simple and straight forward pattern for generic services development. They verified that this approach ease the composition of existing services and orchestration new business processes.

VI. Conclusion

In this paper, after an overview of indicator, we described its relevance for enterprise. We tackled its concept and how to construct it. We are inspired by COBIT 5 processes which cover all of the process normally found in an enterprise relating to IT activities. We also presented COBIT 5 process capability model. We introduced SOA approach in order to improve alignment of business and IT. SOA is an architectural approach based on several principles.

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For future research direction, we will identify the link between SOA and COBIT processes. We will represent how services/processes communicate and cooperate for effective and efficient use of IT. We will release our own indicators, determine these which represent the KPIs, and design our indicator framework architecture.

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